ASSOCIATED DETAIL SPECIFICATION

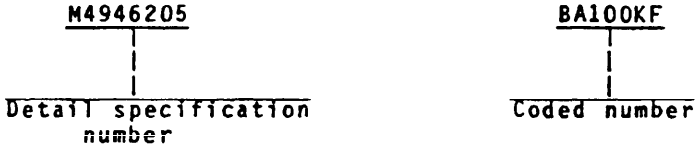
RESISTORS, FIXED, FILM, HIGH VOLTAGE, STYLES RHV50, RHV51, RHV52, RHV53, RHV54, RHV55, RHV56, RHV57

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RHV50, RHV51, RHV52, RHV53, RHV54, RHV55, RHV56, and RHV57 resistors.

1.2 Part or Identifying Number (PIN). Resistors covered by this specification shall be identified by a PIN which shall consist of the basic number of this specification and a coded number. The PIN shall be in the following form:



The coded number shall be derived in accordance with MIL-R-49462..

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-R-39032 - Resistors, Packaging of.
MIL-R-49462 - Resistors, Fixed, Film, High Voltage, General Specification For.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order or precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

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3. REQUIREMENTS

3.1 Associated detail specifications. The individual item requirements shall be in accordance with MIL-R-49462, and as specified herein.

3.2 Design and construction. The resistor shall be of the design and construction, and physical dimensions specified on figure 1.

3.3 Power rating. The power rating shall be as specified in table I, based on a full load operation at an ambient temperature of +70°C. For operation at temperatures higher than +70°C, derate in accordance with figure 2.

TABLE I. Power rating.

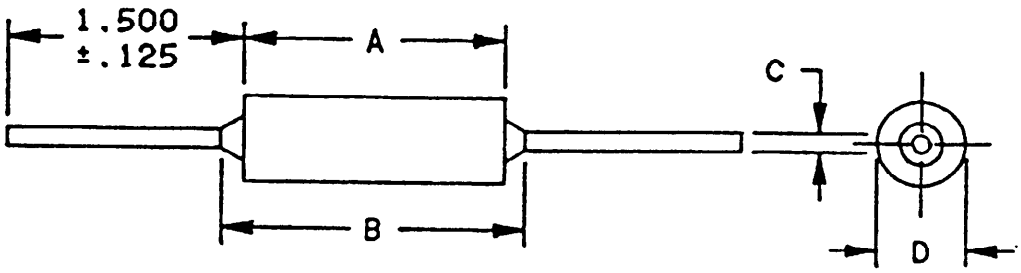
Resistor style	Power rating (watts)	Style
RHV50	0.75	A
RHV51	1.00	B
RHV52	1.50	C
RHV53	2.00	D
RHV54	4.00	E
RHV55	7.00	F
RHV56	10.00	G
RHV57	12.00	H

3.4 Resistance. The minimum and maximum nominal resistance values shall be as specified in table II.

TABLE II. Minimum and maximum nominal resistance values.

Resistor style	Resistance value		Style
	Minimum	Maximum	
RHV50	10 kilohms	5 gigohms	A
RHV51	10 kilohms	10 gigohms	B
RHV52	10 kilohms	20 gigohms	C
RHV53	10 kilohms	100 gigohms	D
RHV54	10 kilohms	5 gigohms	E
RHV55	10 kilohms	10 gigohms	F
RHV56	10 kilohms	15 gigohms	G
RHV57	10 kilohms	20 gigohms	H

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	Dimensions							
	RHV50	RHV51	RHV52	RHV53	RHV54	RHV55	RHV56	RHV57
A max	.460 (11.68)	.500 (12.70)	.750 (19.05)	1.500 (38.10)	1.125 (28.57)	2.125 (53.97)	3.125 (79.37)	4.125 (104.77)
B max	.530 (13.46)	.570 (14.47)	.820 (20.82)	1.584 (40.23)	1.209 (30.70)	2.209 (56.10)	3.209 (81.50)	4.209 (106.90)
C	.020 ±.002 (0.50) ±(0.05)	.032 ±.002 (0.81) ±(0.05)	.032 ±.002 (0.81) ±(0.05)	.032 ±.002 (0.81) ±(0.05)	.032 ±.002 (0.81) ±(0.05)	.032 ±.002 (0.81) ±(0.05)	.032 ±.002 (0.81) ±(0.05)	.032 ±.002 (0.81) ±(0.05)
D max	.145 (3.68)	.170 (4.31)	.170 (4.31)	.170 (4.31)	.325 (8.25)	.325 (8.25)	.325 (8.25)	.325 (8.25)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Millimeters are in parenthesis.

FIGURE 1. Resistor, fixed, film, high voltage.

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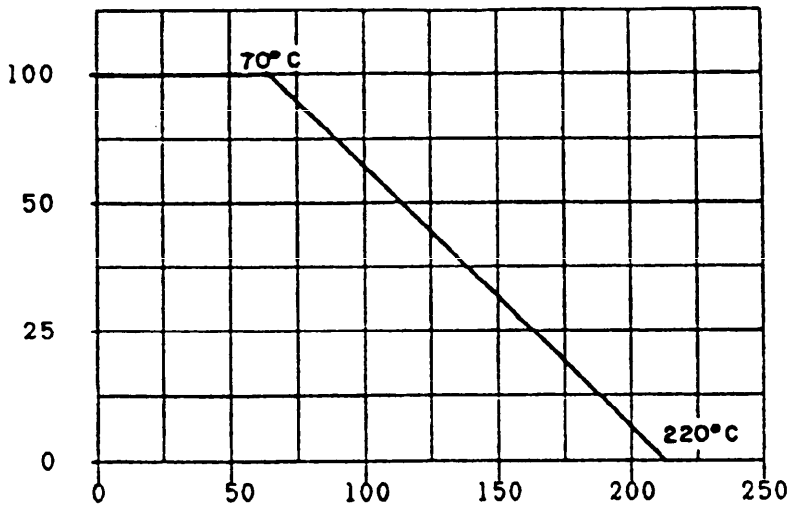


FIGURE 2. Derating curve.

3.5 Maximum voltages. Maximum voltages shall be in accordance with table III.

TABLE III. Maximum voltages.

Resistor style	Voltage (volts maximum kV)	Style
RHV50	2.0	A
RHV51	2.5	B
RHV52	5.0	C
RHV53	10.0	D
RHV54	8.0	E
RHV55	16.0	F
RHV56	24.0	G
RHV57	32.0	H

3.6 Resistance tolerance. The resistance tolerance is identified by a single letter in accordance with table IV.

TABLE IV. Resistance tolerance.

Symbol	Resistance tolerance percent (±)
F	1.0
G	2.0
J	5.0

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-49462.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39032.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-R-49462 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR
Navy - AS, OS
Air Force - 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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